PCBs Enhance Collagen I Expression from Human Peritoneal Fibroblasts

Objective To test the effect of four polychlorinated biphenyl congeners (PCB-77, PCB-105, PCB 153 and PCB 180) on expression of three adhesion markers, TGF-β1, VEGF, and type I collagen in normal human peritoneal and adhesion fibroblasts Design Cell culture study Settings University Research Laboratory Patients Primary cultures of normal peritoneal and adhesion fibroblasts were established from three patients. Interventions Fibroblasts were treated with PCB-77, PCB-105, PCB-153 or PCB-180, 20 ppm for 24 hours. Total RNA was extracted from each treatment and subjected to real-time RT/PCR. Main Outcome and Measures mRNA levels of type I collagen, VEGF and TGF-β1. Results Normal human peritoneal fibroblasts expressed type I collagen, VEGF and TGF-β1. Exposure of normal human fibroblasts to PCB-77, PCB-105, PCB-153 or PCB-180 did not affect mRNA levels of β-actin, the house keeping gene used to normalized RNA levels for the real-time RT/PCR, nor did it affect cell viability as assessed by trypan blue exclusion. PCB treatments, compared to control, resulted in no significant change for TGF-β1 or VEGF mRNA levels, in normal peritoneal and adhesion fibroblasts. In marked contrast, type I collagen mRNA levels were markedly increased in response to the brief 24 hrs exposure to each PCB, treatment each ($P<0.0001$) in both cell types. Conclusion The finding that PCB-77, PCB-105, PCB-153 or PCB-180 increased the expression of type I collagen in human normal peritoneal and adhesion fibroblasts is the first demonstration of involvement of organochlorines in the pathogenesis of tissue fibrosis. This may implicate organochlorine exposure as an etiologic factor in a wide variety of previously unlinked human ailments characterized by fibrosis.